DAFFODIL INTERNATIONAL UNIVERSITY

Department of Software Engineering

ACADEMIC RESEARCH PAPER

AI & MACHINE LEARNING EVALUATION METRICS

A Comparative Study of AI/ML Evaluation Metrics Using Simulated Results

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Abstract

Evaluation metrics are essential for determining whether an artificial intelligence (AI) or machine learning (ML) system performs adequately for its intended task, data distribution, error costs, and deployment environment. This study organizes evaluation into classification, regression, clustering, ranking/recommendation, generative/large language model (LLM) evaluation, and operational evaluation, while also emphasizing threshold selection, calibration, and cost-sensitive evaluation. This study follows that structure and introduces a new set of illustrative numerical results so that the examples differ from the source material.

A simulated evaluation framework is used to demonstrate how complementary metrics change the interpretation of model quality. For classification, the experiment uses a synthetic customer-churn scenario with 1,200 observations and compares two hypothetical classifiers. For regression, clustering, ranking, LLM evaluation, and operational performance, separate simulated result sets are presented. The results demonstrate that no single metric is sufficient: accuracy can conceal class imbalance, RMSE can expose large errors more strongly than MAE, clustering scores depend on geometry and reference labels, ranking metrics emphasize position, and LLM systems require factual and task-oriented evaluation in addition to text overlap. The study concludes that metric selection should follow the ML task, data characteristics, error costs, model objective, and deployment constraints.

Keywords: machine learning evaluation, classification metrics, regression metrics, clustering, ranking, LLM evaluation, calibration, robustness, latency, model performance

1. Introduction

AI/ML systems can be evaluated in fundamentally different ways depending on what the system produces. A classifier may be judged by correct and incorrect class decisions; a regression model by numerical prediction error; a clustering algorithm by internal separation or agreement with reference labels; a ranking system by the quality and order of retrieved items; and a generative system by correctness, factual consistency, task completion, or human judgment. Operational characteristics such as latency, throughput, memory, and energy consumption can also determine whether a technically accurate model is practical to deploy.

This study considers six major metric families—classification, regression, clustering, ranking/recommendation, generative/LLM systems, and operational evaluation—and emphasizes that metric choice must reflect class balance, error cost, and deployment requirements. This revised paper retains those families while changing the numerical examples and adding a coherent simulated evaluation study.

The purpose of the study is not to identify a universally superior metric. Instead, it demonstrates how different metrics answer different evaluation questions. A model may have strong overall accuracy but weak minority-class recall; two regressors may have similar MAE but different RMSE; and an LLM may produce fluent text while making unsupported claims. Therefore, the research approach treats metric selection as an application-dependent decision.

2. Research Objectives

- Explain the major AI/ML evaluation metrics covered in previous research paper.
- Show how classification, regression, clustering, ranking, generative/LLM, and operational metrics address different evaluation goals.
- Replace the source paper's numerical examples with a new set of illustrative values for academic demonstration.
- Demonstrate the effect of class imbalance, error magnitude, ranking position, confidence calibration, and deployment conditions on evaluation results.
- Provide practical guidance for selecting complementary metrics and designing a reliable evaluation protocol.

3. Scope of the Study

The study follows the metric scope in previous research: Accuracy, Balanced Accuracy, Precision, Recall/Sensitivity, Specificity, F1, MCC, ROC-AUC, PR-AUC/Average Precision, Log Loss, and Fβ for classification; MAE, MSE, RMSE, MAPE, MSLE, R^2, and Explained Variance for regression; Silhouette, Calinski–Harabasz, Davies–Bouldin, ARI, and NMI for clustering; Precision@k, Recall@k, MAP, NDCG, and MRR for ranking; Exact Match, BLEU/ROUGE, Groundedness, Faithfulness, Task Success, and Human Evaluation for generative/LLM systems; and latency, throughput, memory/energy use, calibration, and robustness for operational evaluation. The supplied source establishes these families and explains that metric choice should follow the task and failure modes. □filecite□turn0file0□L63-L103□

All new numerical results below are simulated for demonstration. They should be replaced by actual measurements if this paper is submitted as an empirical research study.

4. Research Methodology

A comparative evaluation approach was used to examine how different AI and machine learning evaluation metrics reflect model performance across different tasks. The study considers classification, regression, clustering, ranking and recommendation, generative/LLM systems, and operational performance.

For each evaluation area, appropriate metrics were selected based on the type of prediction, the characteristics of the data, and the performance aspect being measured. Simulated datasets and hypothetical model results were used to demonstrate how different metrics can produce different interpretations of model quality.

For classification, a synthetic customer-churn dataset containing 1,200 observations was considered. Two hypothetical classifiers were evaluated using a confusion matrix and several complementary metrics, including accuracy, balanced accuracy, precision, recall, specificity, F1-score, and MCC.

For regression, two hypothetical forecasting models were compared using MAE, MSE, RMSE,

MAPE, R^2, and explained variance. The comparison was designed to demonstrate the difference between typical prediction error and the effect of larger errors.

For clustering, two configurations were evaluated using both internal and external metrics. Silhouette, Calinski–Harabasz, and Davies–Bouldin were used to examine cluster structure, while ARI and NMI were used to measure agreement with reference labels.

For ranking and recommendation, Precision@5, Recall@5, MAP, NDCG@5, and MRR were used to examine both retrieval coverage and the position of relevant results. Generative/LLM performance was evaluated using Exact Match, ROUGE-1, groundedness, faithfulness, task success, and human evaluation.

Finally, operational performance was examined using latency, throughput, memory usage, energy consumption, calibration error, and robustness retention. All numerical results presented in the study are simulated for academic demonstration and are not claimed to represent measurements from a real-world deployed system.

4.1 Classification Experiment

A synthetic customer-churn dataset of 1,200 observations is assumed. The class distribution contains 960 non-churn customers (80%) and 240 churn customers (20%). The data are divided into 70% training, 15% validation, and 15% test sets. Preprocessing is assumed to be fitted only on the training partition. Two hypothetical classifiers, Model A and Model B, are evaluated on the held-out test set.

Dataset component	Count	Percentage
Total observations	1,200	100%
Non-churn	960	80%
Churn	240	20%
Training	840	70%
Validation	180	15%
Test	180	15%

4.2 Regression Experiment

A simulated demand-forecasting problem is used to compare two hypothetical regression models over 300 test observations. The simulated examples deliberately include a small number of larger errors so that the difference between MAE and RMSE can be illustrated.

4.3 Clustering Experiment

Two simulated embedding-clustering configurations are evaluated using internal and external metrics. The first configuration uses four clusters and the second uses five clusters. Reference labels

are available for ARI and NMI, while Silhouette, Calinski–Harabasz, and Davies–Bouldin provide internal evaluation.

4.4 Ranking and LLM Evaluation

A synthetic search/recommendation benchmark is used for Precision@k, Recall@k, MAP, NDCG, and MRR. A separate 100-response LLM evaluation set is used for Exact Match, groundedness, faithfulness, task success, and human evaluation. These numbers are simulated and are included only to demonstrate metric interpretation.

4.5 Operational Evaluation

The deployment experiment assumes a real-time inference service measured under a fixed workload. Latency, throughput, peak memory, energy per 1,000 requests, calibration error, and robustness retention are reported as illustrative values.

5. Evaluation Metrics Overview

Metric family	Metrics	Primary evaluation focus
Classification	Accuracy, Balanced Accuracy, Precision, Recall, Specificity, F1, MCC, ROC-AUC, PR-AUC/AP, Log Loss, Fβ	Class decisions and probability/ranking quality
Regression	MAE, MSE, RMSE, MAPE, MSLE, R^2, Explained Variance	Numerical prediction error and explained variation
Clustering	Silhouette, Calinski–Harabasz, Davies–Bouldin, ARI, NMI	Cluster cohesion/separation or agreement with labels
Ranking / Recommendation	Precision@k, Recall@k, MAP, NDCG, MRR	Relevance and ordering of retrieved items
Generative / LLM	Exact Match, BLEU/ROUGE, Groundedness, Faithfulness, Task Success, Human Evaluation	Correctness, overlap, factual support, task completion, human judgment
Operational	Latency, Throughput, Memory/Energy, Calibration, Robustness	Deployment responsiveness, capacity, resource use, confidence quality, resilience

6. Classification Metrics

Classification metrics evaluate discrete class predictions. The confusion matrix provides the foundation for accuracy, precision, recall, specificity, F1, and MCC. The analysis considers that the same classifier can look different depending on which metric is used, particularly under class imbalance. □filecite□turn0file0□L104-L140□

Metric	Formula / method	Purpose
Accuracy	(TP+TN)/(TP+TN+FP+FN)	Overall proportion of correct predictions.
Balanced Accuracy	Average class recall	Useful when class frequencies are unequal.
Precision	TP/(TP+FP)	Quality of positive predictions.
Recall / Sensitivity	TP/(TP+FN)	Coverage of actual positive cases.
Specificity	TN/(TN+FP)	Correct rejection of negative cases.
F1	2PR/(P+R)	Harmonic mean of precision and recall.
MCC	Correlation of predicted and actual labels	Uses all four confusion-matrix categories.
ROC-AUC	Area under ROC curve	Threshold-independent discrimination.
PR-AUC / AP	Precision-recall relationship across thresholds	Especially informative for rare positive classes.
Log Loss	Average probabilistic cross-entropy	Evaluates predicted probability quality.
Fβ	$(1+\beta^2)PR/(\beta^2P+R)$	Weights recall and precision differently.

6.1 Classification Results

The following confusion matrices contain newly generated values. They are illustrative rather than observed experimental measurements.

Model	TP	TN	FP	FN
Model A	190	845	115	50
Model B	205	820	140	35

Metric	Model A	Model B
Accuracy	0.863	0.854
Balanced Accuracy	0.836	0.854
Precision	0.623	0.594
Recall	0.792	0.854
Specificity	0.880	0.854
F1	0.697	0.701
MCC	0.617	0.626

The simulated results illustrate a meaningful trade-off. Model B detects more churn cases (higher recall), while Model A produces fewer false-positive alerts (higher precision and specificity). Because the dataset is imbalanced, accuracy alone does not describe this trade-off adequately. A

practical evaluation would therefore report the confusion matrix together with precision, recall, F1, balanced accuracy, and an appropriate threshold-independent measure.

7. Regression Metrics

Regression metrics evaluate continuous predictions. MAE measures average absolute error, MSE squares errors and therefore emphasizes large deviations, RMSE returns the error to the original target scale, MAPE expresses relative error as a percentage, MSLE evaluates logarithmic error for non-negative targets, and R^2 summarizes explained variation. The analysis considers that these measures differ in sensitivity to outliers and scale. □filecite□turn0file0□L371-L399□

Metric	Main interpretation	Typical direction
MAE	Average absolute error in target units	Lower is better
MSE	Average squared error; emphasizes large errors	Lower is better
RMSE	Square root of MSE in target units	Lower is better
MAPE	Average absolute percentage error	Lower is better
MSLE	Squared logarithmic error for non-negative targets	Lower is better
R^2	Variation explained relative to mean baseline	Higher is generally better
Explained Variance	Variance captured by predictions	Higher is generally better

7.1 Regression Results

Model	MAE	MSE	RMSE	MAPE	R^2	Explained Variance
Regression Model A	8.7	176.4	13.28	9.6%	0.872	0.881
Regression Model B	7.9	231.8	15.23	8.8%	0.832	0.846

The simulated values demonstrate why MAE and RMSE should be read together. Model B has the smaller MAE and MAPE, indicating lower typical and relative error, but its larger MSE and RMSE indicate that it contains more severe individual errors. Thus, the choice between models depends on whether typical error or protection against large errors is more important.

8. Clustering Metrics

Clustering evaluation can be internal or external. Internal metrics such as Silhouette, Calinski–Harabasz, and Davies–Bouldin do not require ground-truth labels, whereas ARI and NMI compare predicted partitions with reference labels. The study distinguishes these two evaluation settings. □filecite□turn0file0□L528-L557□

Metric	Interpretation	Direction
Silhouette	Cohesion versus nearest alternative cluster	Higher
Calinski–Harabasz	Between-cluster dispersion relative to within-cluster dispersion	Higher
Davies–Bouldin	Similarity to the most similar neighboring cluster	Lower
ARI	Chance-adjusted agreement with reference partition	Higher
NMI	Normalized information agreement with reference partition	Higher

8.1 Clustering Results

Configuration	Clusters	Silhouette	Calinski–Harabasz	Davies–Bouldin	ARI	NMI
Configuration A	4	0.61	318.5	0.72	0.78	0.82
Configuration B	5	0.57	341.2	0.79	0.84	0.87

The simulated results show that internal and external metrics need not point to exactly the same interpretation. Configuration B has a higher Calinski–Harabasz value and stronger agreement with reference labels, but its Silhouette and Davies–Bouldin values are weaker. This illustrates why clustering studies should avoid selecting a configuration from one metric alone.

9. Ranking and Recommendation Metrics

Ranking metrics evaluate both relevance and position. Precision@k measures the relevant fraction in the visible top-k results, Recall@k measures retrieval completeness within that window, MAP rewards relevant items appearing earlier across queries, NDCG handles graded relevance with rank discounting, and MRR focuses on the position of the first relevant result. The analysis considers that the cutoff k matters because users often inspect only the first few results.
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9.1 Ranking Results

System	Precision@5	Recall@5	MAP	NDCG@5	MRR
Ranker A	0.72	0.48	0.66	0.74	0.71
Ranker B	0.68	0.55	0.63	0.77	0.76

The simulated ranking results illustrate different objectives. Ranker A retrieves a larger proportion of relevant items within the top five, while Ranker B produces stronger NDCG and MRR, meaning that high-quality or first useful results appear earlier. A search system should therefore define

whether the primary objective is broad candidate retrieval, graded relevance, or rapid access to the first useful answer.

10. Generative / LLM Evaluation

Generative systems require evaluation beyond surface-form correctness. Exact Match is useful when an answer must exactly match a reference after normalization; BLEU and ROUGE quantify reference overlap; groundedness and faithfulness assess support and consistency with source material; task success evaluates whether a user goal was completed; and human evaluation can capture nuance, usefulness, tone, and semantic quality. The study distinguishes between benchmark overlap metrics and open-ended factual or task-oriented evaluation. □filecite□turn0file0□L770-L821□

10.1 LLM Results

Metric	Simulated result	Interpretation
Exact Match	74%	74 of 100 outputs exactly matched the normalized reference.
ROUGE-1 F1	0.79	Moderate-to-strong lexical overlap in the simulated benchmark.
Groundedness	0.91	Most extracted factual claims were supported by the supplied context.
Faithfulness	0.88	Most evaluated claims remained consistent with source information.
Task Success	86%	86 of 100 simulated sessions reached the predefined goal.
Human Evaluation	4.1 / 5.0	Average rubric-based quality rating across simulated judgments.

The results demonstrate why exact matching should not be treated as the sole LLM metric. A response can fail exact matching while remaining semantically useful, whereas high lexical overlap can coexist with unsupported statements. For source-grounded systems, groundedness and faithfulness provide additional information about factual support.

11. Operational Metrics

Operational evaluation connects model quality with deployment constraints. Latency measures responsiveness, throughput measures completed work per unit time, memory and energy metrics describe resource requirements, calibration measures whether confidence corresponds to observed correctness, and robustness measures performance retention under perturbation. The supplied paper specifically recommends considering these measures when offline quality alone is insufficient. □filecite□turn0file0□L917-L944□

11.1 Operational Results

Measure	Simulated result	Interpretation
Median latency	210 ms	Typical response time.
P95 latency	390 ms	Tail response time for high-latency requests.
Throughput	2,650 tokens/s	Generated tokens per second under the simulated workload.
Peak VRAM	13.8 GB	Maximum GPU memory consumption.
Energy per 1,000 requests	0.42 kWh	Illustrative energy requirement.
ECE	0.064	Average confidence discrepancy under the simulated binning scheme.
Robustness retention	91.5%	Perturbed-test performance retained relative to clean performance.

These figures show that deployment evaluation should not be separated from model quality. A model with strong task metrics may still require optimization if its P95 latency or memory footprint violates application requirements. Similarly, calibration and robustness can expose risks that are invisible in clean test-set accuracy.

12. Thresholds, Calibration, and Cost-Sensitive Evaluation

Many classifiers produce scores or probabilities rather than final labels. A threshold converts those scores into decisions. Changing the threshold changes TP, FP, TN, and FN, and therefore changes precision, recall, F1, and other decision-dependent measures. The study recommends selecting thresholds on validation data and evaluating the finalized model once on a held-out test set. □filecite□turn0file0□L1042-L1064□

Expected misclassification cost can be represented as C(t) = C_FP·FP(t) + C_FN·FN(t). When false negatives are more costly, a lower threshold may be appropriate; when false positives are more costly, a higher threshold may be appropriate. The correct threshold therefore depends on the application and on the quality of probability calibration.

12.1 Threshold Illustration

Threshold	Precision	Recall	F1	False positives	False negatives
0.35	0.61	0.91	0.73	174	22
0.50	0.72	0.79	0.75	115	50
0.65	0.81	0.63	0.71	70	89

The new illustrative values show the expected precision-recall trade-off. Lowering the threshold increases sensitivity but also creates more false-positive alerts. Raising it reduces false positives

while missing more positive cases. Threshold selection should therefore be based on the application's error costs rather than on a default probability such as 0.50.

13. Comparative Analysis of the Results

Evaluation area	Key revised observation	Reason a single metric is insufficient
Classification	Models trade precision against recall.	Class imbalance and unequal FP/FN costs change the preferred operating point.
Regression	Model B has lower MAE but higher RMSE.	Typical error and large-error sensitivity are different properties.
Clustering	Different internal and external metrics favor different configurations.	Cluster geometry and agreement with labels are distinct questions.
Ranking	One ranker has stronger NDCG/MRR while another has stronger Recall@5.	Retrieval completeness and early relevance are different objectives.
LLM	Task success and groundedness exceed exact-match performance.	Semantically valid or source-supported answers may not match a reference string exactly.
Operations	Latency, throughput, resource use, calibration, and robustness vary independently.	Deployment quality includes constraints beyond predictive accuracy.

14. Practical Metric Selection

Metric selection should begin with the decision that the evaluation is intended to support, not with a preference for a familiar score. The study recommends complementary measures because a single number can hide important failure modes. □filecite□turn1file7□

Evaluation pattern	Recommended complementary measures
Balanced binary classification	Confusion matrix + precision + recall + F1 + accuracy for overall context.
Rare-event classification	Class distribution + recall + precision + PR-AUC/AP + MCC or balanced accuracy.
Regression with outliers	MAE + RMSE/MSE to show typical error and large-error sensitivity.
Ranking with graded relevance	NDCG + Precision@k/Recall@k when a fixed visible window matters.
RAG / factual LLM answering	Task correctness or EM where appropriate + groundedness + faithfulness + human evaluation.
Real-time LLM service	Task quality + latency percentiles + throughput + memory/energy + robustness/calibration.

15. Evaluation Best Practices

- Use train/validation/test separation or cross-validation where appropriate.
- Prevent data leakage by fitting preprocessing and feature transformations without using final test data.
- Report class distribution, sample size, data split, and threshold-selection procedure.
- Use confidence intervals or repeated cross-validation when practical.
- Compare against a simple baseline such as a majority-class classifier or mean-prediction regressor.
- Report complementary metrics when one metric can hide an important failure mode.
- Evaluate subgroup performance, robustness, and deployment constraints when relevant.
- Select thresholds on validation data and reserve the held-out test set for final evaluation.
- Document metric definitions and implementation conventions.
- Keep the evaluation dataset and protocol fixed when comparing models.

16. Discussion

The revised simulated results reinforce the central principle of previous research paper: evaluation is task-dependent. In the classification experiment, the minority churn class makes accuracy insufficient by itself. In regression, MAE and RMSE expose different characteristics of the error distribution. In clustering, internal compactness and external agreement can diverge. In ranking, the location of a relevant item affects user-facing quality. In LLM evaluation, exact string matching captures only a narrow definition of correctness, while groundedness, faithfulness, task success, and human evaluation address broader properties.

The operational results further show that model evaluation should continue beyond an offline benchmark. A deployment team may need to trade throughput against latency, monitor resource consumption, assess calibration before using probabilities for decisions, and test robustness against realistic perturbations. Consequently, an evaluation protocol should be designed around the decisions the model will support.

A limitation of this study is that its changed numerical values are simulated rather than collected from a real dataset. Therefore, the results demonstrate methodology and interpretation, not empirical superiority of any real model. For a formal empirical submission, the simulated values should be replaced with reproducible measurements, dataset details, model configurations, confidence intervals, and statistical tests.

17. Conclusion

AI/ML evaluation is not a one-number exercise. Classification, regression, clustering, ranking, generative/LLM, and operational systems produce different kinds of outputs and therefore require different evaluation perspectives. The simulated numerical examples show that complementary metrics reveal information that isolated scores can hide.

For classification, precision, recall, F1, balanced accuracy, MCC, ROC-AUC, PR-AUC, and calibration can provide a fuller picture than accuracy alone. For regression, MAE, RMSE, MAPE, R^2, and related measures answer different questions about error magnitude and explained variation. Clustering requires a distinction between internal structure and agreement with reference labels. Ranking evaluation must consider both relevance and position. LLM evaluation should combine task correctness with factual support and, where appropriate, human judgment. Operational measures connect offline model quality to real deployment constraints.

The main methodological conclusion is that metric selection should follow the ML task, dataset characteristics, class balance, error costs, model objective, and deployment requirements. The changed data in this paper are intentionally illustrative; actual research should replace them with transparent, reproducible measurements.

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Appendix A. Summary of Simulated Evaluation Results

This appendix summarizes the newly introduced values so that they can be distinguished from the source paper's examples. All values are simulated for academic demonstration.

Area	Changed values introduced
Classification	TP/TN/FP/FN for Models A and B; derived accuracy, balanced accuracy, precision, recall, specificity, F1, MCC.
Regression	MAE, MSE, RMSE, MAPE, R^2, and explained variance for two hypothetical models.
Clustering	Four- and five-cluster Silhouette, CH, DB, ARI, and NMI values.
Ranking	Precision@5, Recall@5, MAP, NDCG@5, and MRR for two hypothetical rankers.
LLM	Exact Match, ROUGE-1 F1, groundedness, faithfulness, task success, and human rating.
Operational	Median/P95 latency, throughput, VRAM, energy, ECE, and robustness retention.
Threshold analysis	Three operating thresholds with revised precision, recall, F1, FP, and FN values.

Appendix B. Source-Basis Note

The supplied paper is the basis for the terminology, organization, metric families, methodological recommendations, and reference list used in this revised document. The source itself states that it covers classification, regression, clustering, ranking/recommendation, generative/LLM, operational evaluation, and threshold/calibration/cost-sensitive evaluation. □filecite□turn0file0□L19-L36□

The numerical values in this revised paper were intentionally changed and are explicitly identified as simulated. This avoids representing invented results as observations from the original research.